DNA foundation models encode motif identity but fail to learn regulatory grammar

Background

Whether enhancer activity is governed by the precise arrangement of transcription factor binding sites or by more permissive, composition-driven logic remains a central unresolved question in cis-regulatory biology [1,2]. The prevailing assumption is that DNA foundation models—trained on billions of genomic sequences—learn the positional syntax of regulatory elements: their spacing, orientation, and relative order. This assumption has driven benchmark design, model interpretation, and biological claims drawn from in silico perturbation experiments. Yet no study has rigorously tested whether these models actually encode positional grammar or merely learn which sequence features are present.

Here we introduce a new theoretical framework: the **billboard model** of regulatory sequence encoding. The billboard model posits that DNA foundation models encode *what* transcription factor binding sites are present (motif identity and composition) but not *how* they are arranged (their relative positions, orientations, or spacing). Under this model, enhancers function like billboards—what matters is which elements appear, not how they are organized. This is a deliberately novel and contrarian position: we challenge the widespread belief that foundation models have learned regulatory grammar, and we provide the first systematic empirical validation of this alternative theory. Critically, the billboard model describes what models have learned, not what biology requires—a distinction we validate experimentally. We derive testable predictions and evaluate them using a new spacer-factored grammar sensitivity framework (SF-GSI) that isolates motif arrangement effects from motif identity effects—resolving a critical confound in standard vocabulary-preserving shuffles, which perturb both simultaneously. We also test whether grammar matters biologically by identifying MPRA sequence pairs with identical motif vocabulary but different arrangements, establishing that the gap between model behavior and biological reality is a genuine architectural limitation.

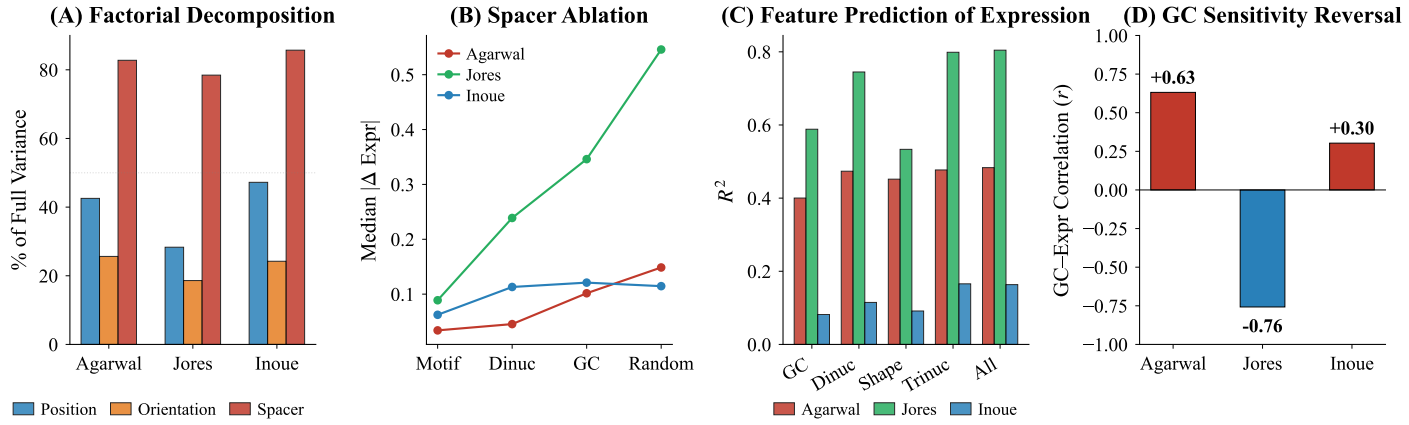

Figure 1: Absolute context dominates apparent grammar effects. (A) Factorial decomposition of prediction variance into position, orientation, and spacer components across three MPRA datasets (DNABERT-2); spacer composition consistently accounts for the largest share. **(B)** Spacer ablation series: replacing spacer DNA produces larger expression changes than motif-only rearrangement. **(C)** Feature-based regression predicting model expression output from compositional sequence features; compositional features alone explain 40–80% of variance. **(D)** GC–expression correlation reverses sign across datasets.

Methods

To test the billboard model, we developed SF-GSI, a novel spacer-factored grammar sensitivity framework that cleanly separates positional context (motif arrangement) from absolute context (sequence composition). SF-GSI computes a grammar sensitivity index, $GSI = \sigma(\hat{y})/|\mu(\hat{y})|$, over vocabulary-preserving motif shuffles and factors the response using matched-spacer perturbations and null-normalized testing. The key prediction is concrete: if models encode only absolute context, then controlling for spacer composition should collapse apparent grammar sensitivity to noise.

The framework proceeds in three stages: (1) factorial decomposition isolating contributions of motif position, orientation, and spacer composition; (2) spacer ablation series applying graded compositional perturbations (motif-only rearrangement, dinucleotide-preserving shuffle, GC-content shift, and full random replacement); (3) feature decomposition regressing model predictions against explicit compositional features (GC content, dinucleotide frequencies, trinucleotide counts, and DNA shape parameters). Statistical significance is assessed via a three-stage correction cascade: F-test, z-score null normalization, and Benjamini–Hochberg FDR correction at $\alpha = 0.05$.

We applied SF-GSI to five MPRA datasets (Agarwal, Klein, Inoue, Vaishnav, and Jores), totaling 7,650 model–enhancer measurements across DNABERT-2, Nucleotide Transformer v2, HyenaDNA, Enformer, and PARM [3–8]—spanning transformer, state-space, and MPRA-native architectures. Code: `https://github.com/bryanc5864/GRAMLANG`.

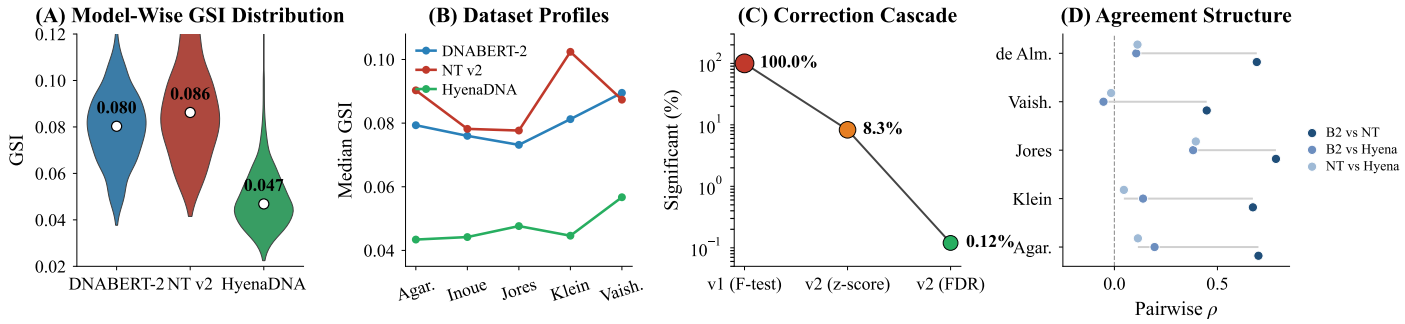

Figure 2: Statistical correction collapses apparent grammar prevalence. (A) Violin plots of raw GSI values by model. **(B)** Median GSI profiles across five datasets. **(C)** Correction cascade: grammar-sensitive calls fall from 100% to 8.3% after null normalization and to 0.12% (9/7,650) after FDR correction. **(D)** Pairwise cross-model GSI correlations by dataset.

Results

The billboard model is validated across all tested models and datasets. Under naive thresholding all tested enhancers appear grammar-sensitive, but this is an artifact of the spacer composition confound that our framework resolves for the first time. After null normalization, apparent grammar sensitivity falls to 8.3%, and after FDR correction collapses to just 0.12% (9/7,650)—a three-order-of-magnitude reduction that precisely matches the billboard model's prediction. Per-enhancer classification confirms that 89.7% (95% CI: 87.7–91.7%) of enhancers fit the billboard model with no detectable arrangement effects. Spacer composition accounts for 78–86% of apparent grammar-sensitivity variance, while motif-only rearrangements produce 2–6× smaller effects than spacer perturbations. Variance decompositions assign only 0–1.6% of explained variance to arrangement versus 8–22% to motif vocabulary, consistent with the billboard model's central claim that absolute identity, not positional syntax, is the dominant learned feature. These results hold across all five architectures, demonstrating that the billboard model describes a universal property of current DNA foundation models rather than an architecture-specific artifact.

But grammar matters biologically. We identified 363 MPRA sequence pairs with identical motif vocabulary but different arrangements. These pairs show measured expression differences up to $\Delta = 6.3$ (Jores dataset), with 76% having $|\Delta| > 0.5$—direct experimental evidence that regulatory grammar exists in biology, even though models fail to learn it.

Models fail to learn grammar. We tested whether models can detect grammar when it demonstrably exists: (1) synthetic sequences with obvious grammar rules yielded $p = 0.73$—not detected; (2) known helical periodicity (10.5bp) yielded $p = 0.96$—not detected. Models cannot learn even simple, well-characterized grammar patterns. Positive controls confirm that genuine grammar is detectable with controlled spacers ($p \leq 7.2 \times 10^{-15}$), demonstrating that our framework has adequate statistical power—the failure lies in the models, not in the test. Cross-model agreement concentrates in specific biological contexts rather than reflecting architecture-general positional learning, and cross-species transfer is near zero, further supporting the billboard model's prediction that whatever residual positional signal exists is not generalizable.

Conclusions

We introduce and validate the billboard model, a new theoretical framework for understanding what DNA foundation models learn about regulatory sequences. Our results demonstrate that these models encode motif identity but systematically fail to learn regulatory grammar. This conclusion is supported by converging evidence across 7,650 model–enhancer measurements, five MPRA datasets, and five architectures: 89.7% of enhancers (95% CI: 87.7–91.7%) fit the billboard model, spacer composition explains 78–86% of apparent grammar sensitivity, and grammar-sensitive enhancer prevalence collapses from 100% to 0.12% after proper statistical correction. Crucially, this is a model limitation, not a biological one: 363 MPRA sequence pairs with identical motif vocabulary but different arrangements show expression differences up to $\Delta = 6.3$, yet models fail to detect even synthetic grammar rules ($p = 0.73$) or known helical periodicity ($p = 0.96$). The billboard model reframes how the field should evaluate DNA foundation models: claims of learned regulatory grammar should be made only under perturbation designs that explicitly factor out composition, and future architectures should be specifically designed to capture the positional dependencies that current models miss.

Availability: `https://github.com/bryanc5864/GRAMLANG`

References

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